

AI Co-Pilot: Territory & Quota (T&Q) System Architecture

1. Executive Summary

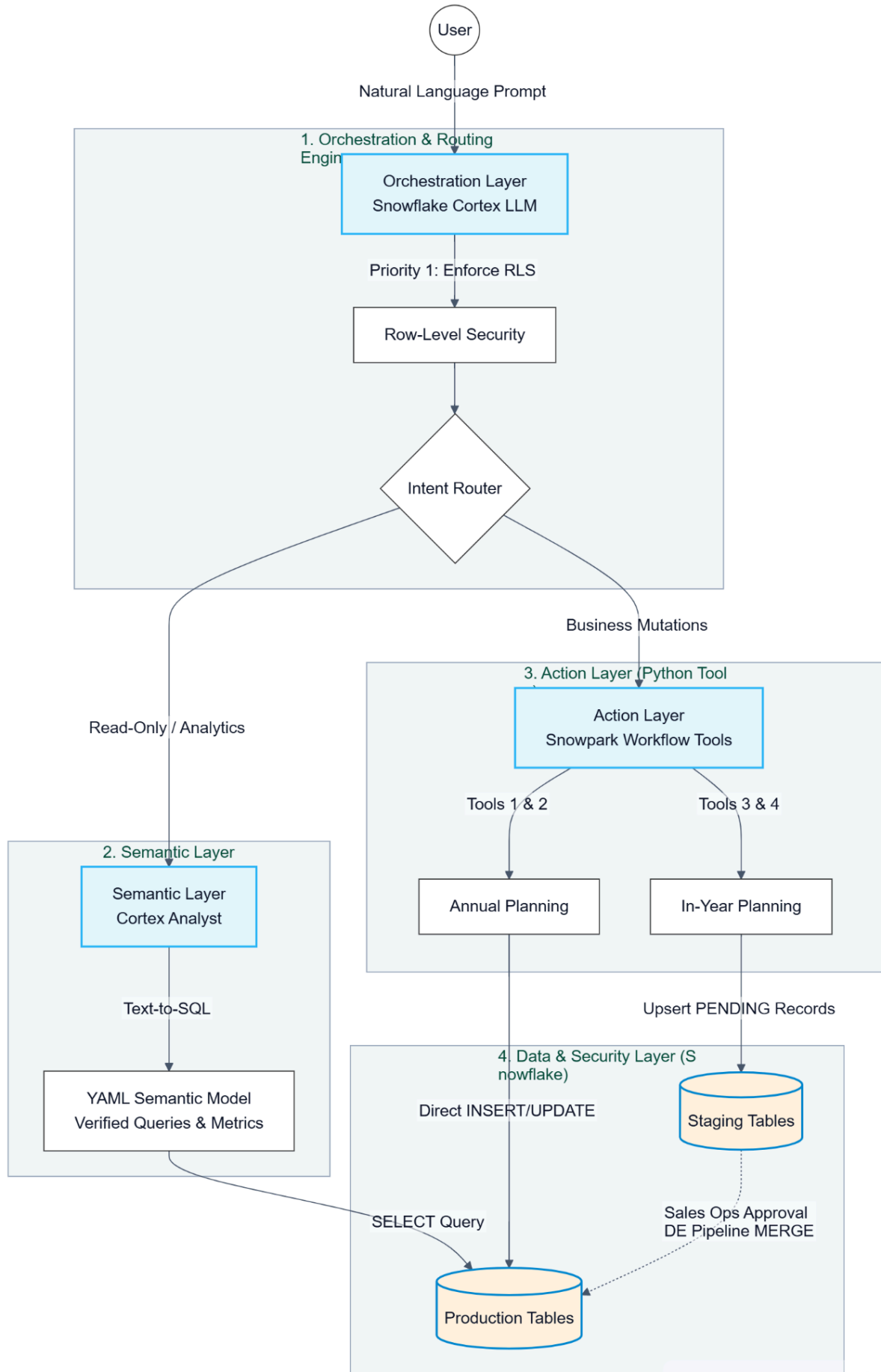
The AI Co-Pilot is an enterprise-grade Snowflake AI Agent designed to orchestrate **Territory and Quota planning**. It bridges the gap between conversational AI and rigid financial systems by utilizing a dual-engine approach: **Cortex Analyst (Text-to-Sql Engine)** for robust semantic reporting, and **Custom Python Functions (Tools)** for secure, governed database mutations.

2. High-Level System Architecture

The architecture is divided into four distinct layers to isolate read-only analytics from state-changing workflows.

- **Orchestration Layer (The LLM):** The core decision engine. It uses strict, ranked routing instructions to determine whether a user intent requires a database mutation, a semantic query, or a system audit.
- **Action Layer (Workflow Tools):** A suite of Python-based Snowpark functions that execute business logic (e.g., BoB Assignments, Quota Updates). These tools enforce fail-fast validations and generate audit trails before any data is written.
- **Semantic Layer (Cortex Analyst):** A highly structured YAML semantic model built over the analytics views. It handles all descriptive reporting (e.g., "Top 10 customers," "YoY Growth") using predefined metrics, dynamic division logic, and verified SQL queries.
- **Data & Security Layer (Snowflake):** The underlying transaction tables protected by native Row-Level Security (RLS/RBAC) to ensure users only interact with their permitted territories.

3. Architectural Flow & Data Routing



4. The Dual-Planning Engine (Annual vs. In-Year)

The agent dynamically adapts its data writing strategy based on the active planning cycle (Annual vs. In-Year).

Annual Planning Workflow

During the Annual Planning cycle (for the whole year), the database acts as a clean slate.

- **Execution:** Changes are written **directly to production tables**
- **Audit:** Every change is logged synchronously into an audit table.

In-Year Planning Workflow

Mid-year changes require strict governance to preserve historical revenue attribution and active quotas.

- **Staging Environment:** In-Year changes never touch production directly. All updates and assignments are written to a staging table
- **Approval Pipeline:** Staged records are flagged as **PENDING**. Once Sales Ops approves the request (handled in the agent as well), a Data Engineering pipeline merges the staged data into the production quota table.

5. Security & Governance Engine

To prevent LLM hallucinations during critical financial updates, the agent operates on a strict governance model.

- **Row-Level Security (RLS / RBAC):** All queries dynamically scope to **the user who is accessing the chatbot**. The agent is strictly forbidden from manually filtering by any ID when a user asks for "my team".
- **Workflow Isolation:** If a user intent contains mutation verbs (e.g., "update," "drop," "assign"), the agent is strictly forbidden from using the read-only Cortex Analyst. It must route directly to the designated Workflow Tool.
- **Preview-First Execution:** All business mutations follow a **PREVIEW -> CONFIRM -> EXECUTE** pattern. The database is never updated without explicit user confirmation of the preview payload.